Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)	(i)		<u>12 J</u> transferred [from chemical to electrical or work done] (1) per {unit charge / coulomb} [passing through cell] (1)	2			2		
		(ii)		pd drops below emf **or** energy is wasted inside cell **or** the internal resistance limits the current [that the battery can deliver]		1		1		
	(b)	(i)	I.	7.0 V across parallel combination **or** $R = 20\ \Omega \times \frac{0.35\ \text{A}}{0.25\ \text{A}}$ **or** by implication (1) $R = \left[\frac{7.0\ \text{V}}{0.25\ \text{A}}\right] = 28\ [\Omega]$ (1)		2		2	1	
			II.	5.0 V across r **ecf or** resistance of parallel combi = 11.7 Ω **ecf and** total resistance in circuit = 20.0 Ω **ecf or** by implication (1) $r = \left[\frac{5.0\ \text{V}}{0.60\ \text{A}}\right] = 8.3\ [\Omega]$ (1)		2		2	1	
		(ii)	I.	$0.35^2 \times 20$ (1) $\times\ 40 \times 60$ [= 5.9 kJ or 5880 J] (1) **Alternative:** 0.35 × 7 **ecf** (1) × 40 × 60 [= 5.9 kJ or 5880 J] (1) **Alternative:** $\frac{7^2\ \textbf{ecf}}{20}$ (1) × 40 × 60 [= 5.9 kJ or 5880 J] (1)	1	1		2	2	
			II.	From electrical (accept KE of electrons) into random **or** vibrational **or** thermal **or** internal (accept heat) (1) [Free] electrons collide with {ions / atoms / lattice} (1)	2			2		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(b)	(iii)		{Parallel combination / total circuit} resistance increases **or** main current decreases (1) So greater [proportion of 12] V across parallel combination **or** smaller pd across r **or** pd across battery terminals increases (1) So for 20 Ω resistor, pd is greater **or** current is greater, so Dafydd is correct (1) No argument that doesn't involve r is valid, but give 1 mark for greater [proportion than before of] current goes through the 20 Ω			3	3		
				Question 3 total	**5**	**6**	**3**	**14**	**4**	**0**